# Residual conductivity at isotropic Dirac points

**Abstract**

We consider spinless, noninteracting electrons in a two-dimensional real periodic potential. Suppose that the Fermi energy meets the Bloch spectrum only at finitely many isolated, isotropic, two-band conical crossings. We prove that the current induced by a suddenly applied constant electric field, first linearized in the field strength and then averaged in time, has conductivity $M/16$ times the identity, where $M$ is the number of inequivalent crossings. The proof explains the independence from the cone velocities: time averaging selects momenta on the inverse-time scale around each crossing, and a change of scale cancels the velocity factors. The conical energy assumptions also determine the required current matrix elements; no separate assumption on Bloch eigenvectors is needed.

## 1 The result and its mechanism

When the Fermi energy meets the bands only at isolated Dirac points, there is no Fermi curve. A weak electric field can still drive transitions between occupied and empty states across arbitrarily small gaps. We prove that these transitions leave a finite contribution after a long time average, and identify why each isotropic cone contributes the same constant.

We work in atomic units and count one spin species. The theorem is conditional on the band structure specified below; deriving that structure from a particular lattice potential is a separate question.

### 1.1 Model and assumptions

Let $\mathcal{R} \subset \mathbb{R}^2$ be a lattice with fundamental cell $\Omega$. Its reciprocal lattice is

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\},$$

and $\mathcal{B}$ denotes a fundamental cell of $\mathcal{R}^*$. Our convention gives $|\Omega|\,|\mathcal{B}| = (2\pi)^2$. For a real potential $V \in H^1_{\mathrm{per}}(\Omega)$, set

$$H = -\tfrac{1}{2}\Delta + V, \qquad H_k = \tfrac{1}{2}(-i\nabla + k)^2 + V. \tag{1.1}$$

The fibers act on $L^2_{\mathrm{per}}(\Omega;\mathbb{C})$, have common domain $H^2_{\mathrm{per}}(\Omega;\mathbb{C})$, and have compact resolvent. Write their eigenvalues, with multiplicities, as

$$\lambda_1(k) \le \lambda_2(k) \le \cdots, \qquad \lambda_0(k) = -\infty.$$

All statements about momentum are understood modulo $\mathcal{R}^*$.

Fix a Fermi energy $\mu_{\mathrm{F}}$ and an integer $N \ge 1$. Assume that there are finitely many distinct momenta $k_1, \dots, k_M$ such that

$$\lambda_N(k) < \mu_{\mathrm{F}} < \lambda_{N+1}(k) \qquad \text{if } k \notin \{k_1, \dots, k_M\}, \tag{1.2}$$

and that, at every $k_i$,

$$\lambda_{N-1}(k_i) < \lambda_N(k_i) = \mu_{\mathrm{F}} = \lambda_{N+1}(k_i) < \lambda_{N+2}(k_i). \tag{1.3}$$

Finally, for some $v_i > 0$,

$$\begin{aligned} \lambda_N(k_i + q) &= \mu_{\mathrm{F}} - v_i|q| + O(|q|^2), \\ \lambda_{N+1}(k_i + q) &= \mu_{\mathrm{F}} + v_i|q| + O(|q|^2) \end{aligned} \qquad (q \longrightarrow 0). \tag{1.4}$$

The slopes may differ between crossings. The remainders are uniform in the direction of $q$.

Let $e_\beta, e_\alpha \in \mathbb{R}^2$ be the field and current directions; they need not be unit vectors. For $\varepsilon \geq 0$, define

$$\begin{gathered} \gamma_0 = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H), \qquad H_\varepsilon = H - \varepsilon x \cdot e_\beta, \qquad J_\alpha = (-i\nabla) \cdot e_\alpha, \\ \gamma_\varepsilon(t) = e^{-itH_\varepsilon}\gamma_0 e^{itH_\varepsilon}, \qquad j_\varepsilon(t) = \underline{\mathrm{Tr}}\big(J_\alpha \gamma_\varepsilon(t)\big). \end{gathered} \tag{1.5}$$

Here $H_\varepsilon$ is the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2)$, and the trace per unit area is

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}(\mathbf{1}_\Omega A \mathbf{1}_\Omega) \tag{1.6}$$

for periodic, locally trace-class operators. The operator facts needed to make these definitions valid at the stated regularity are proved in Appendix A.

**Theorem 1.1.** *Under* (1.2)–(1.4)*, the current in* (1.5) *is well defined for every* $\varepsilon \geq 0$ *and* $t \geq 0$*. For every fixed* $t$*, the linear response*

$$\sigma_{\alpha\beta}(t) = \lim_{\varepsilon \downarrow 0} \frac{j_\varepsilon(t)}{\varepsilon}$$

*exists, and*

$$\boxed{\lim_{T\to\infty} \frac{1}{T} \int_0^T \sigma_{\alpha\beta}(t)\, dt = \frac{M}{16}\, e_\alpha \cdot e_\beta.} \tag{1.7}$$

The order of operations is part of the statement: the field strength tends to zero at each fixed time, and the resulting response is then averaged over time. For two inequivalent cones the spinless coefficient is $1/8$. No spin multiplicity is included in $M$.

## 1.2 How the proof works

The field transports crystal momentum. Thus the response is governed by the change of the occupied subspace as $k$ moves. Although that subspace has no continuous extension through a cone, its first derivative grows only as $|k - k_i|^{-1}$, which is integrable in two dimensions. This is enough to differentiate the exact evolution and obtain an interband formula for the current.

For an occupied–empty pair with energy gap $\Delta > 0$, time averaging replaces the oscillatory response factor by

$$\frac{1}{T} \int_0^T \frac{\sin(t\Delta)}{\Delta^2}\, dt = \frac{1 - \cos(T\Delta)}{T\Delta^3}. \tag{1.8}$$

Every uniformly positive gap therefore contributes $O(T^{-1})$. Near the $i$th cone, $\Delta \sim 2v_i|q|$, so the surviving scale is $|q| \sim (v_i T)^{-1}$. The current matrix elements supply a factor $v_i^2$, which cancels in this rescaling. What remains is an angular average and the scalar integral $\int_0^\infty (1 - \cos s) s^{-2}\, ds = \pi/2$.

We first derive the response in Section 2, then determine the current matrix elements from the conical energies in Section 3. The rescaling that completes the proof occupies Section 4.

## 2 The occupied subspace and the finite-time response

Use the fiber velocities and directional derivatives

$$v_\nu(k) = (-i\nabla + k)\cdot e_\nu, \qquad \partial_\nu = e_\nu \cdot \nabla_k \quad (\nu = \alpha, \beta),$$

and write

$$P(k) = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H_k). \tag{2.1}$$

Away from the crossings, $P(k)$ has rank $N$. Its value at a crossing does not affect a direct integral or any formula below, so we may change it there when discussing regularity. We use tr for the fiber trace and $\|\cdot\|_1$ for the trace norm.

With the Bloch normalization used here, a bounded periodic operator with integrable fiber trace norm satisfies

$$\underline{\mathrm{Tr}}(A) = \frac{1}{(2\pi)^2}\int_{\mathcal{B}} \operatorname{tr} A(k)\, dk. \tag{2.2}$$

The normalization and local trace-class assertion are justified in Appendix A.2.

### 2.1 Regularity across a crossing

Individual eigenvectors need not be smooth at a degeneracy. An isolated group of bands, however, has an analytic spectral projection. Near a Dirac point we therefore keep the two touching bands together, choose a smooth frame for their joint subspace, and only then distinguish the occupied line from the empty one.

**Proposition 2.1** (Regularity of the occupied subspace)**.** *The projection $P$ is real analytic away from the crossings and has integrable weak first derivatives with values in the trace class. Near each $k_i$,*

$$\|P(k)\|_1 \le C, \qquad \|\partial_\beta P(k)\|_1 \le \frac{C}{|k-k_i|}. \tag{2.3}$$

*There is no point-supported term in its weak derivative. Moreover the same estimates hold with $P$ and $\partial_\beta P$ replaced by $\Lambda P\Lambda$ and $\Lambda(\partial_\beta P)\Lambda$, where $\Lambda = 1 - \Delta_{\mathrm{per}}$.*

*Proof.* The fibers form an analytic family on the common domain $H^2_{\mathrm{per}}$. The Riesz projection of any isolated finite group of bands is consequently analytic, including as a map into this domain; see [2, §XII.2] and Appendix A.1. This proves the assertion away from the crossings.

Fix one crossing, write $k = k_i + q$, and let $Q(q)$ be the analytic rank-two projection onto its touching pair. If $W_0 : \mathbb{C}^2 \to \operatorname{Ran} Q(0)$ is an isometry, then, for $q$ small,

$$W(q) = Q(q)W_0\big(W_0^* Q(q) W_0\big)^{-1/2} \tag{2.4}$$

is an analytic orthonormal frame. Its columns and their derivatives lie in $H^2_{\mathrm{per}}$. The reduced matrix

$$h(q) = W(q)^* H_{k_i+q} W(q) \tag{2.5}$$

has precisely the two touching eigenvalues. Write

$$h(q) = a(q)I + b(q), \qquad \operatorname{tr} b(q) = 0, \qquad d(q) = \sqrt{\tfrac{1}{2}\operatorname{tr} b(q)^2}.$$

Its eigenvalues are $a(q) \pm d(q)$, and (1.4) gives $d(q) = v_i|q| + O(|q|^2)$. Since $b(0) = 0$ and $b$ is analytic, the lower eigenprojection satisfies

$$\pi_-(q) = \tfrac{1}{2}\big(I - b(q)/d(q)\big), \qquad \|\partial_\beta \pi_-(q)\| \le C|q|^{-1}. \tag{2.6}$$

Let $P_<(q)$ be the analytic projection onto the $N-1$ bands below the pair (zero if $N=1$). Then

$$P(k_i+q) = P_<(q) + W(q)\pi_-(q)W(q)^*. \tag{2.7}$$

This proves (2.3). It also proves the weighted version: after multiplication by $\Lambda$ on both sides, the formula is a finite sum of rank-one operators formed from $H^2$ vectors and their derivatives.

Finally, integrate by parts with disks of radius $\rho$ around the crossings removed. The boundary terms are $O(\rho)$, because $P$ is bounded in the stated norms and each boundary circle has length $O(\rho)$. They tend to zero, while $|q|^{-1}$ is locally integrable. Thus the derivatives computed off the crossings are the full weak derivatives. Reciprocal-cell boundaries are identified by the usual Bloch unitaries, so this is also a global statement on the momentum torus. □

The Sobolev weights in the proposition serve one purpose: they allow the derivative of the density to be paired with the unbounded velocity. The geometric content is the integrable $1/|q|$ singularity.

## 2.2 Transport by the electric field

In Bloch coordinates, multiplication by $x \cdot e_\beta$ becomes $i\partial_\beta$. The evolved density remains periodic in space; its Bloch fibers $\rho_\varepsilon(t,k)$ satisfy

$$\partial_t \rho_\varepsilon + \varepsilon \partial_\beta \rho_\varepsilon = -i[H_k, \rho_\varepsilon], \qquad \rho_\varepsilon(0,k) = P(k). \tag{2.8}$$

The transport term expresses the acceleration of crystal momentum. To differentiate in $\varepsilon$, it is useful to integrate this equation exactly before taking a limit.

For fixed final time $t$ and momentum $k$, the characteristic is

$$k_s = k - \varepsilon(t-s)e_\beta, \qquad 0 \le s \le t.$$

Let $U_\varepsilon(t,s;k)$ be the propagator from $s$ to $t$ for $r \mapsto H_{k_r}$. Transport from the initial momentum gives

$$\rho_\varepsilon(t,k) = U_\varepsilon(t,0;k)P(k-\varepsilon t e_\beta)U_\varepsilon(t,0;k)^*. \tag{2.9}$$

For almost every final $k$, this segment avoids the crossings, so $P$ is smooth along it. Since $[H_{k_s}, P(k_s)] = 0$, differentiation gives

$$\frac{d}{ds}\big(U_\varepsilon(t,s;k)P(k_s)U_\varepsilon(t,s;k)^*\big) = \varepsilon U_\varepsilon(t,s;k)(\partial_\beta P)(k_s)U_\varepsilon(t,s;k)^*.$$

Integrating from 0 to $t$ yields the exact identity

$$\frac{\rho_\varepsilon(t,k) - P(k)}{\varepsilon} = -\int_0^t U_\varepsilon(t,s;k)(\partial_\beta P)(k_s)U_\varepsilon(t,s;k)^*\, ds. \tag{2.10}$$

This formula separates the two effects of the field: it shifts the occupied subspace and propagates the resulting change.

As $\varepsilon \downarrow 0$, the propagator tends to $e^{-i(t-s)H_k}$, and translations of $\partial_\beta P$ converge in $L^1$. The weighted version of Proposition 2.1 justifies this limit inside the current trace; the estimates are given in Appendix A.3. Consequently

$$\sigma_{\alpha\beta}(t) = -\frac{1}{(2\pi)^2}\int_{\mathcal{B}}\int_0^t \operatorname{tr}\Big(v_\alpha(k)e^{-isH_k}(\partial_\beta P)(k)e^{isH_k}\Big)\, ds\, dk. \tag{2.11}$$

Here the equilibrium current is zero. Indeed, complex conjugation maps $H_k$ to $H_{-k}$ and $v_\alpha(k)$ to $-v_\alpha(-k)$, so $\operatorname{tr}(v_\alpha P)$ is odd on the momentum torus.

### 2.3 The interband formula

Choose a measurable orthonormal eigenbasis $u_n(k)$ of $H_k$. No smooth choice is needed. For an occupied index $m \le N$ and an empty index $n > N$, set

$$\begin{aligned} \Delta_{mn}(k) &= \lambda_n(k) - \lambda_m(k), \\ Z_{mn}(k) &= \langle u_m, v_\alpha(k) u_n \rangle \langle u_n, v_\beta(k) u_m \rangle. \end{aligned} \tag{2.12}$$

Inner products are conjugate linear in the first argument. The sums below are independent of the basis choices within degenerate eigenspaces.

**Proposition 2.2** (Finite-time response)**.** *For every fixed $t \ge 0$, the response exists and is*

$$\sigma_{\alpha\beta}(t) = \frac{2}{(2\pi)^2} \int_{\mathcal{B}} \sum_{m \le N < n} \operatorname{Re} Z_{mn}(k) \frac{\sin(t\Delta_{mn}(k))}{\Delta_{mn}(k)^2} \, dk. \tag{2.13}$$

*The sum and integral are absolutely convergent.*

*Proof.* Differentiating $P^2 = P$ shows that $\partial_\beta P$ has only occupied–empty and empty–occupied blocks. Differentiating $[H_k, P(k)] = 0$ then gives

$$\langle u_n, (\partial_\beta P) u_m \rangle = -\frac{\langle u_n, v_\beta u_m \rangle}{\Delta_{mn}} \qquad (m \le N < n). \tag{2.14}$$

Thus the variation of a completely filled band subspace creates interband coherence. There is no diagonal block or additional derivative concentrated at a crossing.

Substitution in (2.11), followed by the elementary time integral, gives

$$\frac{2}{(2\pi)^2} \int_{\mathcal{B}} \sum_{m \le N < n} \frac{\operatorname{Re} Z_{mn} \sin(t\Delta_{mn}) + \operatorname{Im} Z_{mn} (1 - \cos(t\Delta_{mn}))}{\Delta_{mn}^2} \, dk. \tag{2.15}$$

Time reversal leaves the gaps unchanged and conjugates $Z_{mn}$: both velocity matrix elements change sign, so their product acquires no sign. The imaginary part in (2.15) therefore integrates to zero, proving (2.13). Periodicity makes this argument independent of the chosen reciprocal cell.

For convergence, completeness and Cauchy–Schwarz give the uniform bound

$$\sum_{m \le N < n} |Z_{mn}(k)| \le \left( \sum_{m \le N} \|v_\alpha(k) u_m\|^2 \right)^{1/2} \left( \sum_{m \le N} \|v_\beta(k) u_m\|^2 \right)^{1/2} \le C. \tag{2.16}$$

The last inequality follows from the uniform graph bound for the occupied states. In particular, no estimate on high-energy eigenfunctions is required. The smallest occupied–empty gap is bounded below away from the crossings and is comparable to $|k - k_i|$ near $k_i$. Since

$$\frac{|\sin(t\Delta)|}{\Delta^2} \le \frac{t}{\Delta}, \qquad \frac{1 - \cos(t\Delta)}{\Delta^2} \le \frac{t^2}{2},$$

both (2.15) and (2.13) are absolutely integrable. These bounds also justify the preceding spectral expansion and time integration. □

## 3 What an isotropic cone determines

The response formula involves eigenvectors through $Z_{mn}$, whereas our assumptions specify only eigenvalues. The missing link is the geometry of a two-dimensional eigenspace: isotropic conical eigenvalues force its linearized Hamiltonian to satisfy a rigid matrix identity.

Fix one crossing $k_i$, put $v = v_i$, and use the frame $W(q)$ and reduced matrix $h(q)$ from (2.4)–(2.5). For $q = r\omega$, where $r > 0$ and $|\omega| = 1$, let $P_-(q)$ and $P_+(q)$ denote the rank-one projections onto bands $N$ and $N+1$ at $k_i + q$.

**Lemma 3.1** (Current matrix elements at a cone)**.** *As $q = r\omega \to 0$, uniformly in $\omega$,*

$$\operatorname{Re}\operatorname{tr}(P_-(q)v_\alpha(k_i+q)P_+(q)v_\beta(k_i+q)) = v^2\big[e_\alpha\cdot e_\beta - (e_\alpha\cdot\omega)(e_\beta\cdot\omega)\big] + O(r). \tag{3.1}$$

*Proof.* Expand the analytic matrix at the crossing:

$$h(q) = \mu_{\mathrm{F}} I + A(q) + O(r^2), \qquad A(q) = q_1A_1 + q_2A_2. \tag{3.2}$$

For every $\omega$, the eigenvalues of $A(\omega)$ are $-v$ and $v$, by (1.4). Homogeneity therefore gives

$$\operatorname{tr} A(q) = 0, \qquad A(q)^2 = v^2|q|^2 I, \qquad A(a)A(b) + A(b)A(a) = 2v^2(a\cdot b)I \tag{3.3}$$

for all $a, b \in \mathbb{R}^2$; the last identity follows by polarization. In particular the limiting lower and upper projections are

$$R_\pm(\omega) = \tfrac{1}{2}(I \pm A(\omega)/v), \qquad W(q)^*P_\pm(q)W(q) = R_\pm(\omega) + O(r). \tag{3.4}$$

The error estimate also follows directly from (2.6).

The physical velocity has the corresponding leading matrix:

$$W(q)^*v_\alpha(k_i+q)W(q) = A(e_\alpha) + O(r). \tag{3.5}$$

To see this without assuming a fixed frame, differentiate $h = W^*HW$ at $q = 0$. The terms differentiating $W$ sum to $\mu_{\mathrm{F}}\partial_\alpha(W^*W) = 0$, while the remaining term is $W_0^*v_\alpha(k_i)W_0$. This equals $\partial_\alpha h(0) = A(e_\alpha)$, and analyticity proves (3.5).

It remains to compute a two-by-two trace. For $R = R_-(\omega)$, the identities in (3.3) give

$$\operatorname{tr}(RA(a)) = -v(a\cdot\omega), \qquad \operatorname{Re}\operatorname{tr}(RA(a)A(b)) = v^2(a\cdot b).$$

Since $R$ has rank one,

$$\begin{aligned}\operatorname{Re}\operatorname{tr}\big(RA(a)(I-R)A(b)\big) &= \operatorname{Re}\operatorname{tr}(RA(a)A(b)) - \operatorname{tr}(RA(a))\operatorname{tr}(RA(b))\\ &= v^2\big[a\cdot b - (a\cdot\omega)(b\cdot\omega)\big].\end{aligned}$$

Combining this with (3.4) and (3.5) proves (3.1). □

The bracket in (3.1) pairs the components of the two directions perpendicular to $\omega$. For a longitudinal response, it is the squared tangential component of the field direction. For the linearized cone, the radial velocity acts within each band, while the tangential velocity couples the two bands. This explains the angular factor in the final step.

## 4 Time averaging and the universal constant

We now prove Theorem 1.1. The finite-time assertions follow from Proposition 2.2 and Appendix A. It remains to take the time average. Absolute convergence on every bounded time interval gives

$$\frac{1}{T}\int_0^T \sigma_{\alpha\beta}(t)\,dt = \frac{2}{(2\pi)^2}\int_{\mathcal{B}} \sum_{m\le N<n} \operatorname{Re} Z_{mn}(k)\,\frac{1-\cos(T\Delta_{mn}(k))}{T\Delta_{mn}(k)^3}\,dk. \tag{4.1}$$

### 4.1 Only the touching pair survives

Choose disjoint small disks around the crossings on the momentum torus. Outside these disks, all occupied–empty gaps have a common positive lower bound. Inside each disk, the same is true for every pair other than $(N, N+1)$, by (1.3). Thus (2.16) and $|1-\cos s| \le 2$ show that the total contribution of these transitions is $O(T^{-1})$.

Only the touching pair in each disk remains.

### 4.2 One cone

Suppress its index and write

$$\Delta(q) = \lambda_{N+1}(k_i+q) - \lambda_N(k_i+q), \qquad w(q) = \operatorname{Re} Z_{N,N+1}(k_i+q).$$

For $q = r\omega$, (1.4) and Lemma 3.1 give

$$\begin{gathered} \Delta(r\omega) = 2vr + O(r^2), \qquad w(r\omega) = v^2 g(\omega) + O(r), \\ g(\omega) = e_\alpha \cdot e_\beta - (e_\alpha \cdot \omega)(e_\beta \cdot \omega). \end{gathered} \tag{4.2}$$

Take the disk radius $\delta$ small enough that $cr \le \Delta(r\omega) \le Cr$. Its contribution to (4.1) is

$$I_T = \frac{2}{(2\pi)^2} \int_0^{2\pi} \int_0^{\delta} w(r\omega) \frac{1-\cos(T\Delta(r\omega))}{T\Delta(r\omega)^3}\, r\, dr\, d\theta, \qquad \omega = (\cos\theta, \sin\theta). \tag{4.3}$$

Set $s = 2vTr$ and, only for this change of scale, write $D_T(s,\omega) = T\Delta(s\omega/(2vT))$. Then

$$I_T = \frac{1}{8\pi^2 v^2} \int_0^{2\pi} \int_0^{2vT\delta} w\Big(\frac{s\omega}{2vT}\Big) \frac{s\big(1-\cos D_T(s,\omega)\big)}{D_T(s,\omega)^3}\, ds\, d\theta. \tag{4.4}$$

For fixed $s > 0$ and $\omega$, $D_T(s,\omega) \to s$ and $w(s\omega/(2vT)) \to v^2 g(\omega)$. This is the step at which the velocity cancels: the $v^2$ in the transition strength cancels the $v^{-2}$ introduced by the radial change of scale.

To justify the limit, extend the integrand in (4.4) by zero beyond $2vT\delta$. The gap bounds imply that $D_T$ is comparable to $s$ throughout its support. Since $w$ is bounded and $1-\cos x \le \min\{x^2/2, 2\}$, the absolute value of this integrand is bounded by

$$C \min\{1, s^{-2}\}. \tag{4.5}$$

This bound is integrable and independent of $T$ and $\omega$. Dominated convergence therefore yields

$$\lim_{T\to\infty} I_T = \frac{1}{8\pi^2} \left( \int_0^{2\pi} g(\omega)\, d\theta \right) \left( \int_0^{\infty} \frac{1-\cos s}{s^2}\, ds \right). \tag{4.6}$$

Thus only the linearized cone contributes to the limit.

The angular integral is

$$\int_0^{2\pi} g(\omega)\, d\theta = \pi\, e_\alpha \cdot e_\beta.$$

For the radial integral, the positive Laplace representation of $s^{-2}$ and Tonelli's theorem give the short evaluation

$$\int_0^{\infty} \frac{1-\cos s}{s^2}\, ds = \int_0^{\infty} u \left( \frac{1}{u} - \frac{u}{u^2+1} \right) du = \int_0^{\infty} \frac{du}{1+u^2} = \frac{\pi}{2}.$$

Substitution in (4.6) gives $\lim_{T\to\infty} I_T = (e_\alpha \cdot e_\beta)/16$. Summing over the $M$ disks and adding the vanishing gapped contribution proves (1.7). □

# A Operator framework and the current limit

This appendix verifies the analytic facts used in the main argument for $V \in H^1_{\mathrm{per}}(\Omega)$. The only external operator results used are the standard self-adjointness, Bloch decomposition, analytic perturbation, and common-domain evolution results in Reed–Simon.

## A.1 Self-adjointness, graph bounds, and evolution

A periodic $L^2$ potential in two dimensions is uniformly locally $L^2$ and is infinitesimally bounded relative to $-\Delta$; see [2, Theorem XIII.96]. Thus, on the plane and on the periodic cell, for every $\eta > 0$,

$$\|Vu\|_2 \le \eta\|\Delta u\|_2 + C_\eta\|u\|_2. \tag{A.1}$$

The corresponding form bound

$$\int |V|\,|u|^2 \le \eta\|\nabla u\|_2^2 + C_\eta\|u\|_2^2 \tag{A.2}$$

follows from the two-dimensional Sobolev interpolation inequality on cells and a partition of unity. Kato–Rellich [1, Theorem X.12] gives the domains asserted in (1.1); the fiber graph norms are uniformly equivalent to the $H^2$ norm when $k$ ranges over a bounded set. Compactness of the cell gives compact resolvent. These observations also show that $k \mapsto H_k$ is analytic from $H^2_{\mathrm{per}}$ to $L^2_{\mathrm{per}}$ and that its resolvents, hence its finite-rank Riesz projections, are analytic into $H^2_{\mathrm{per}}$; compare [2, §XII.2].

For the Stark operator, apply the Faris–Lavine theorem [1, Theorem X.38] to

$$2H_\varepsilon = -\Delta + V_1 + W, \qquad V_1(x) = -2\varepsilon x \cdot e_\beta, \qquad W(x) = 2V(x).$$

Here $V_1 \ge -c|x|^2 - d$ for suitable $c, d > 0$, and $-a\Delta + W$ is bounded below for any $a > 0$ by (A.2). The auxiliary operator $-\Delta + V_1 + W + 2c|x|^2$ is essentially self-adjoint on $C_c^\infty$: the linear term and $W$ are infinitesimally bounded relative to the harmonic oscillator, by (A.1) and the oscillator graph estimate for $\Delta$. Kato–Rellich applies. The space $C_c^\infty$ is invariant under coordinates and derivatives, as required in Faris–Lavine. Its conclusion is exactly the essential self-adjointness used in (1.5).

For a continuously differentiable momentum path $k(r)$ in a bounded set, the Hamiltonians $H_{k(r)}$ have common domain $H^2_{\mathrm{per}}$ and smoothly varying coefficients as operators from that domain to $L^2_{\mathrm{per}}$. The common-domain evolution theorem [1, Theorem X.70] therefore supplies their unitary propagators and bounded propagation in the common graph norm. Indeed, the theorem's resolvent and difference-quotient hypotheses follow immediately from

$$H_{k+h} - H_k = h \cdot (-i\nabla + k) + \tfrac{1}{2}|h|^2 \tag{A.3}$$

and the uniform graph bounds, after a fixed scalar shift of the generator. In particular, with $\Lambda = 1 - \Delta_{\mathrm{per}}$,

$$\|\Lambda U_\varepsilon(t, s; k)\Lambda^{-1}\| \le C_t \qquad (0 \le s \le t, 0 \le \varepsilon \le 1), \tag{A.4}$$

uniformly for $k$ in a fixed reciprocal cell. For arbitrary fixed $\varepsilon$ and finite $t$, the analogous bound still holds.

## A.2 Bloch normalization and local traces

On a dense class of functions use the Bloch transform

$$(\mathcal{U}f)(k,x) = \sum_{R\in\mathcal{R}} e^{-ik\cdot(x+R)} f(x+R), \qquad x \in \Omega, \tag{A.5}$$

with direct-integral measure $dk/|\mathcal{B}|$. It is unitary, sends $H$ to the fibers $H_k$, and sends $J_\alpha$ to $v_\alpha(k)$; see [2, Theorems XIII.97 and XIII.99]. The reciprocal identifications are

$$f(k+K,x) = e^{-iK\cdot x} f(k,x), \qquad H_{k+K} = e^{-iK\cdot x} H_k e^{iK\cdot x}.$$

The projections, velocities, and characteristic propagators obey the same covariance. Multiplication by $e^{-iK\cdot x}$ preserves $H^2_{\mathrm{per}}$, so the weighted estimates remain valid on every bounded enlargement of a cell. We use this extension when translating momentum across a cell boundary.

Here is a direct verification of (2.2). Let $E : L^2(\Omega) \to L^2(\mathbb{R}^2)$ extend by zero. Then $(\mathcal{U}Ef)(k,x) = e^{-ik\cdot x} f(x)$, so for a bounded periodic operator $A$ with integrable fiber trace norm,

$$E^* A E = \frac{1}{|\mathcal{B}|} \int_{\mathcal{B}} e^{ik\cdot x} A(k) e^{-ik\cdot x}\, dk. \tag{A.6}$$

The right side is a Bochner integral in the trace class. Taking its trace and dividing by $|\Omega|$ gives (2.2). Blocks between two translated cells have the same integral with an additional scalar phase of modulus one, and are also trace class. A bounded spatial set meets only finitely many cells, so its compression is a finite sum of such blocks. This proves local trace class.

The field evolution is also described by (A.5). Translation in momentum followed by the fiber propagator defines a unitary group for $t \in \mathbb{R}$,

$$f(k) \longmapsto U_\varepsilon(t,0;k) f(k - \varepsilon t e_\beta).$$

Concatenating the momentum characteristics gives the group law, and the continuity of propagators and translations gives strong continuity. Differentiating at $t = 0$ on Bloch transforms of $C_c^\infty(\mathbb{R}^2)$ shows that its self-adjoint generator extends $H_k - i\varepsilon\partial_\beta$, the transform of $H_\varepsilon$. Essential self-adjointness therefore identifies this group with $e^{-itH_\varepsilon}$. Conjugating the decomposable projection $P$ gives (2.9); in particular $\gamma_\varepsilon(t)$ is periodic.

On bounded momentum sets, the occupied projection has uniformly bounded range in $H^2$. Equation (A.4) shows that, on finite time intervals, the same is true of the rank-$N$ projection $\rho_\varepsilon(t,k)$. Thus $v_\alpha(k)\rho_\varepsilon(t,k)$ is bounded and has uniformly bounded trace norm. Consequently $J_\alpha\gamma_\varepsilon(t)$ is a bounded, periodic, locally trace-class operator, and its current is given by (2.2).

## A.3 Passing to the derivative inside the current

We give the estimates behind the passage from (2.10) to (2.11). For a fiber operator $A$, use the weighted trace norm

$$\|A\|_{1,2} = \|\Lambda A \Lambda\|_1.$$

Products with $\Lambda$ are understood by their bounded extensions; equivalently $A = \Lambda^{-1} B \Lambda^{-1}$ for a trace-class operator $B$. By Proposition 2.1, $P$ has integrable weak derivatives in this norm. Hence, on bounded momentum sets,

$$\int_{\mathcal{B}} \|(\partial_\beta P)(k-h) - (\partial_\beta P)(k)\|_{1,2}\, dk \longrightarrow 0 \qquad (h \to 0). \tag{A.7}$$

The covariant extension of the field is used whenever $k - h$ leaves the chosen cell.

Write $U_0(t,s;k) = e^{-i(t-s)H_k}$. Duhamel's formula, (A.3), and the graph bounds give

$$\|(U_\varepsilon(t,s;k) - U_0(t,s;k))\Lambda^{-1}\| \le C_t\varepsilon. \tag{A.8}$$

The same difference is uniformly bounded from $L^2$ to $H^2$ after right multiplication by $\Lambda^{-1}$, by (A.4). Interpolation between $L^2$ and $H^2$ therefore yields

$$\|v_\alpha(k)(U_\varepsilon(t,s;k) - U_0(t,s;k))\Lambda^{-1}\| \le C_t\sqrt{\varepsilon}. \tag{A.9}$$

All these bounds are uniform for $0 \le s \le t$ and $k \in \mathcal{B}$.

For $A = \Lambda^{-1}B\Lambda^{-1}$, factor

$$v_\alpha U_\varepsilon A U_\varepsilon^* = (v_\alpha U_\varepsilon \Lambda^{-1}) B (\Lambda^{-1} U_\varepsilon^*).$$

Equations (A.4)–(A.9) give, in trace norm,

$$\begin{aligned} \|v_\alpha U_\varepsilon A U_\varepsilon^*\|_1 &\le C_t \|A\|_{1,2}, \\ \|v_\alpha (U_\varepsilon A U_\varepsilon^* - U_0 A U_0^*)\|_1 &\le C_t\sqrt{\varepsilon}\, \|A\|_{1,2}. \end{aligned} \tag{A.10}$$

Apply the second estimate to $A = (\partial_\beta P)(k)$ and the first to the translated difference in (A.7). Integration over $k$ and $s$ proves convergence of the current traces in (2.10). This establishes (2.11) for every fixed $t$, with no interchange between the small-field and long-time limits.

# References


[1] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, II: Fourier Analysis, Self-Adjointness*, Academic Press, New York, 1975.

[2] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, IV: Analysis of Operators*, Academic Press, New York, 1978.